

Multiple Choice (10 marks)

1. What is leptin resistance?
 - (a) Deficiency of leptin caused by a mutation in the leptin gene
 - (b) Lack of leptin release from the adipose tissue
 - (c) High concentrations of leptin without reduction in energy intake
 - (d) Low concentrations of leptin with increased energy intake
2. Which of the following statements is/are true? How can smoking affect breastfeeding?
 - (a) Suppresses milk production
 - (b) Alters the composition of breast milk
 - (c) Increases the risk of early cessation of breastfeeding
 - (d) all of the options given are correct
3. What is the best method to estimate changes in body fat during weight loss?
 - (a) BMI
 - (b) DXA
 - (c) Four-compartment model
 - (d) MRI
4. Which of the following confirmed values meet the diagnostic threshold for diabetes?
 - (a) fasting blood glucose $\geq$ 140 mg/dl
 - (b) random glucose $>$ 160 mg/dl
 - (c) 2 hour post prandial glucose $\geq$ to 126 mg/dl
 - (d) fasting blood glucose $\geq$ 126 mg/dl
5. Which of the following vitamins is involved in one-carbon metabolism?
 - (a) Folate
 - (b) Riboflavin
 - (c) Thiamin
 - (d) Vitamin C

True / False (10 marks)

Answer True or False

6. True or False: Acetaldehyde is a significant metabolite of alcohol that contributes to cancer development.
7. True or False: Transaminases require a coenzyme derived from vitamin B₆ to facilitate transamination reactions.
8. True or False: Deuterium is a stable isotope commonly used to measure total body water in health sciences.
9. True or False: Food availability and consumption can be measured at five stages within the food supply chain.
10. True or False: Eating only locally grown food in central Africa can lead to iodine deficiency due to limited iodine-rich food sources.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between alcohol consumption and the risk of developing esophageal cancer, including the mechanisms by which alcohol may contribute to this disease.
12. Discuss the role of Vitamin B₆ in the transamination process of amino acids and explain its significance in human metabolism and overall health.

End of Paper